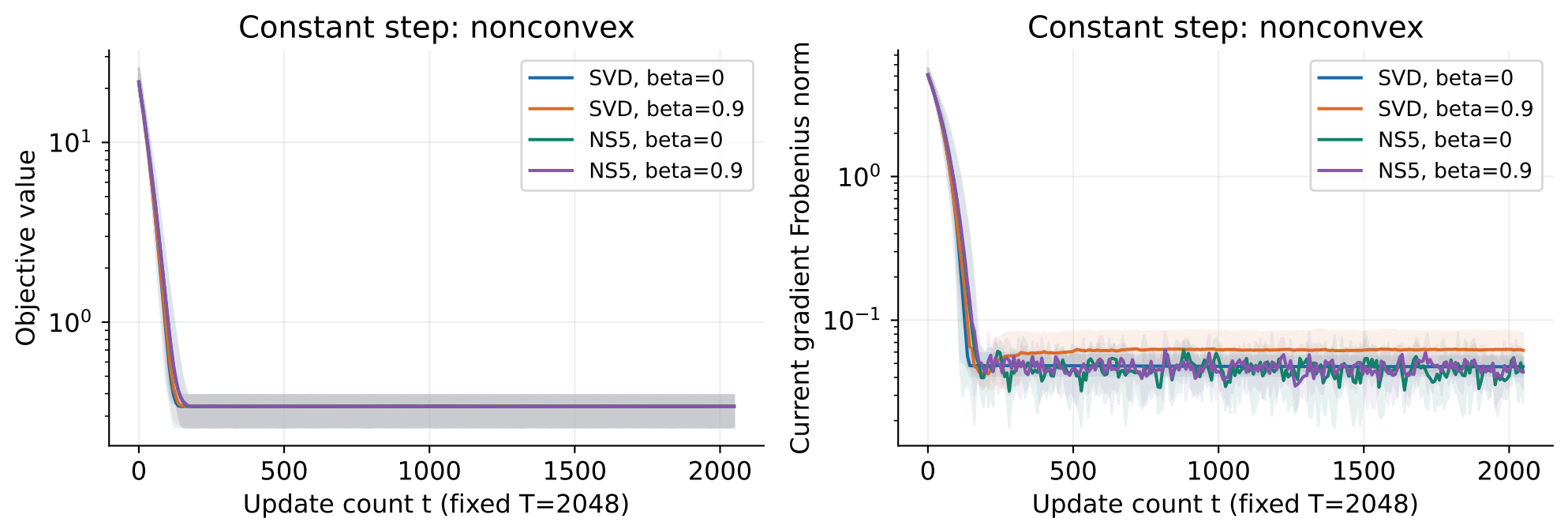
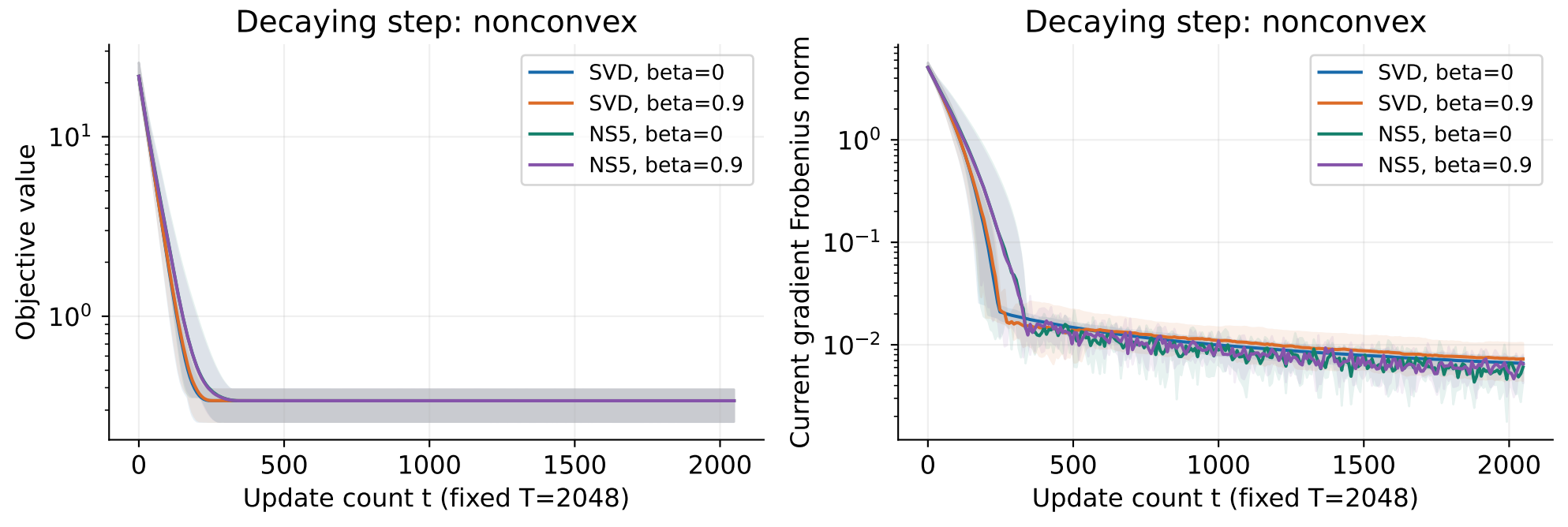

Constant step: nonconvex
Objective value
10^1
10^0
0
500
1000
1500
2000
Update count t (fixed T=2048)
SVD, beta=0
SVD, beta=0.9
NS5, beta=0
NS5, beta=0.9

Constant step: nonconvex
Current gradient Frobenius norm
10^0
10^{-1}
0
500
1000
1500
2000
Update count t (fixed T=2048)
SVD, beta=0
SVD, beta=0.9
NS5, beta=0
NS5, beta=0.9


Decaying step: nonconvex
Objective value
10^1
10^0
0
500
1000
1500
2000
Update count t (fixed T=2048)
SVD, beta=0
SVD, beta=0.9
NS5, beta=0
NS5, beta=0.9

Decaying step: nonconvex
Current gradient Frobenius norm
10^0
10^{-1}
10^{-2}
0
500
1000
1500
2000
Update count t (fixed T=2048)
SVD, beta=0
SVD, beta=0.9
NS5, beta=0
NS5, beta=0.9